

Dr. Olga Soloveva

Postdoctoral researcher

Research interests: [hep-ph, nucl-th, hep-lat]

- QCD matter under extreme conditions: phase transitions and critical phenomena in strongly correlated systems at high T and μ_B , effective models (PNJL, 3D-Ising/CEP mapping), hydrodynamics, and microscopic transport
- Phenomenology of relativistic heavy-ion collisions: thermodynamics, transport coefficients, and microscopic off-shell (PHSD/DQPM) dynamics at finite T and μ_B
- Hard probes of the strongly interacting quark–gluon plasma (sQGP): in-medium parton energy loss (collisional and radiative, LPM), jet substructure, heavy-flavour, and medium response
- Quantum information in QCD: entropy- and entanglement-inspired observables at hadronisation
- Machine learning for heavy-ion physics: interpretable deep learning (CNNs, GNNs, transformers; SHAP) and inverse-problem / surrogate modelling

Current position

Jun 2024 - **Postdoctoral researcher**, GSI Helmholtzzentrum für Schwerionenforschung GmbH,
present Darmstadt, Germany.
Supervisor: Prof. Dr. H. Elfner

Profiles

- Full Name: Olga Evgenyevna Soloveva (Solovjeva in publications 2014–2018)
- ORCID: [0000-0002-8833-4384](https://orcid.org/0000-0002-8833-4384)
- Inspire: [\[Soloveva-Solovjeva\]](#)
- LinkedIn [Olga-Soloveva](#)
- gitlab gitlab.com/osolo
- github github.com/olgasolovev
- website olgasolovev.github.io

Education

May 2018 - **Ph.D. in Physics (Magna Cum Laude)**, Goethe University, Frankfurt am
December Main, (“[Transport properties of the strongly-interacting quark-gluon plasma](#)”);
2021 Supervisor: Prof. Dr. E. Bratkovskaya.

September **Master’s degree in Applied Mathematics and Physics (*with distinction*)**, National
2014 - Research Nuclear University MEPhI (Moscow Engineering Physics Institute), Moscow,
August 2016 Master’s thesis: “Magnetic characteristics of mesons in the lattice gauge theory”;
Supervisors: Dr. E.V. Luschevskaya, Prof. Dr. O.V. Teryaev.

September 2010 - **Bachelor's degree in Applied Mathematics and Physics (*with distinction*)**,
 August 2014 - *National Research Nuclear University MEPhI (Moscow Engineering Physics Institute)*,
 Moscow, Bachelor's thesis: "Determination of the interaction potential between static charges in graphene within Monte-Carlo methods"; Supervisor: Dr. V.V. Braguta.

September 2000 - **Pupil**, *Novosondetskiy Bul'var, 4, Ulyanovsk*, Russia, 432072, email: mou20@uom.mv.ru.
 August 2010

Complementary Education

Apr 2019 **Making an Impact as an effective researcher**, *with the doctoral school HGS-HIRe*, Buchenau, Germany.

Nov 2020 **Leading Teams in a Research Environment**, *with the doctoral school HGS-HIRe*, (online).

Sep 2022 -present **Machine Learning zoomcamp**, *by Alexey Grigorev*, (on Zoom) Berlin, Germany.

11-18 Apr 2023 **The Eighth Machine Learning in HEP summer school**, *by INFN*, Erice, Italy.

Funding and Grants

2022 -2024 HFHF Postdoctoral Scholarship

2018 -2021 Ph.D. Scholarship - Helmholtz Graduate School for Hadron and Ion Research (HGS-HIRe)

2016 Edward Lozansky scholarship for students specialized in theoretical physics.

2015 — Scholarship for students of the federal state educational institutions of professional education in Russian Federation, Government Decree № 945, Nov 18, 2011.

2015 — Scholarship for young scientists (senior students, Post-graduate students and candidates of sciences) within the framework of the "Student Research and Education Program of the FAIR-Russia Research Center"

2017 President Scholarship for graduate students of the federal state educational institutions of professional education in Russian Federation by Government Decree № 1621 of 18.11.2011.

Research experience

Jun 2024 - present **Postdoctoral researcher**, *GSI Helmholtzzentrum für Schwerionenforschung GmbH*, Darmstadt, Germany.
 AG Prof. Dr. Hannah Elfner

Jan 2022 - Jun 2024 **Postdoctoral researcher**, *Helmholtz Research Academy Hesse for FAIR (HFHF) and ITP, Goethe University*, Frankfurt, Germany.
 AG Prof. Dr. Elena Bratkovskaya

2018 - 2021 **Researcher PhD student**, *ITP, Goethe University*, Frankfurt, Germany.
 AG Prof. Dr. Elena Bratkovskaya

2016 - 2018 **Junior researcher**, *Institute for Theoretical and Experimental Physics (ITEP)*, Moscow, Russia.
 Laboratory 191 "Lattice Gauge Theories"

2015 - 2016 **Junior researcher**, *NRNU MEPhI*, Moscow, Russia.
 Department for Theoretical Nuclear Physics

- 2014 - 2015 **Laboratory assistant**, *Institute for Theoretical and Experimental Physics (ITEP)*, Moscow, Russia.
Laboratory 910 "High Energy Physics"
- [Service to the Scientific Community](#)
- Feb 2023 - **member of Thematical Working Group 8 (TWG)**, [sub-group Computing, Machine Learning and Artificial Intelligence](#), NuPECC Long Range Plan 2024.
- present
- 2022 - **referee**, *Physical Review D, Modern Physics Letters A*.
- present
- 17–29 **organizing committee member**, [6th International Conference on New Frontiers in Physics](#), Crete, Greece.
- August 2017

Skills

Languages

- English **Fluent**
- German **Upper - Intermediate (B2)**
- Italian **Basic (A2)**
- Russian **Native**

Technical skills

- Programming languages C++, Fortran, Python, $\text{\LaTeX}$, Bash, HTML/CSS, Wolfram Mathematica
- Operating System Linux, Windows
- HPC SLURM, Version Control System (git)

Publications

41 publications, 557 citations, h-index: 13 (as of June 15, 2026; source: [\[INSPIRE\]](#)) **A complete list of publications can be found under:** [\[INSPIRE: a:"Soloveva, Olga" or a:"Soloveva, O" or a:"Solovjeva, O](#)

White paper/ Invited review

- “Phase Transitions in Particle Physics – Results and Perspectives from Lattice Quantum Chromo-Dynamics”, (Chapter 3.5), Progress in Particle and Nuclear Physics 104070 Vol. 130 (2023) e-print arXiv: 2301.04382 [hep-lat];

Peer-Reviewed Journals

- 1. “ $\pi^\pm$ and $\rho^{0,\pm}$ - mesons in a strong magnetic field on the lattice”
E. V. Luschevskaya, O. A. Kochetkov, O. V. Teryaev, **O.E. Solovjeva**,
JETP Lett. 101 (2015), no. 10, 674-678;
- 2. “Magnetic polarizabilities of light mesons in SU(3) lattice gauge theory”
E. V. Luschevskaya, **O.E. Solovjeva** O. A. Kochetkov, O. V. Teryaev,
Nucl. Phys. B **898** (2015) 627-643; e-print arXiv: 1411.4284 [hep-lat];
- 3. “Magnetic polarizability of pion”
E.V. Luschevskaya, **O.E. Solovjeva**, O.V. Teryaev,
Phys. Lett. B **761** (2016) 393-398; e-print arXiv: 1511.09316 [hep-lat];
- 4. “The Evolution of Meson Masses in a Strong Magnetic Field”
M.A. Andreichikov, B.O. Kerbikov, E.V. Luschevskaya, Yu.A. Simonov,
O.E. Solovjeva,
JHEP **05** (2017) 007; e-print arXiv: 1610.06887 [hep-ph];

- 5. "Determination of the properties of vector mesons in external magnetic field by Quenched $SU(3)$ Lattice QCD"
E.V. Luschevskaya, **O.E. Solovjeva**, O.V. Teryaev,
JHEP **09** (2017) 142; e-print arXiv: 1608.03472 [hep-lat];
- 6. "Tensor polarizability of the vector mesons from $SU(3)$ lattice gauge theory"
E. V. Luschevskaya, O. V. Teryaev, D. Y. Golubkov, **O. Solovjeva** and R. A. Ishkuvatov,
JHEP **11** (2018), 186; e-print arXiv: 1811.02344 [hep-lat];
- 7. "Exploring the partonic phase at finite chemical potential within an extended off-shell transport approach "
P. Moreau, **O. Soloveva**, L. Oliva, T. Song, W. Cassing, E. Bratkovskaya,
Phys. Rev. C **100** (2019) no.1, 014911; e-print arXiv: 1903.10257 [nucl-th];
- 8. "Transport coefficients for the hot quark-gluon plasma at finite chemical potential μ_B "
O. Soloveva, P. Moreau and E. Bratkovskaya,
Phys. Rev. C **101** (2020) no.4, 045203; e-print arXiv:1911.08547 [nucl-th];
- 9. "Exploring the partonic phase at finite chemical potential in and out-of equilibrium"
O. Soloveva, P. Moreau, L. Oliva, V. Voronyuk, V. Kireyeu, T. Song and E. Bratkovskaya,
Particles **3** (2020) no.1, 178-192; e-print arXiv:2001.05395 [nucl-th];
- 10. "Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition"
O. Soloveva, D. Fuseau, J. Aichelin and E. Bratkovskaya,
Phys. Rev. C **103** (2021) no.5, 054901; e-print arXiv:2011.03505 [nucl-th];
- 11. "Diffusion coefficient matrix of the strongly interacting quark-gluon plasma"
J. A. Fotakis, **O. Soloveva**, C. Greiner, O. Kaczmarek and E. Bratkovskaya,
Phys. Rev. D **104** (2021) no.3, 034014; e-print arXiv: 2102.08140 [hep-ph];
- 12. "QCD at finite chemical potential in and out-of-equilibrium"
O. Soloveva, P. Moreau and E. Bratkovskaya,
Phys. Scripta **96** (2021) no.12, 124067; e-print arXiv:2103.15417 [hep-ph] (2021);
- 13. "Transport properties and equation-of-state of hot and dense QGP matter near the critical end-point in the phenomenological dynamical quasi-particle model",
O. Soloveva, J. Aichelin and E. Bratkovskaya,
Phys. Rev. D, **105** (2022) 5, 054011 e-print arXiv:2108.08561 [hep-ph] (2021);
- 14. "Exploring jet transport coefficients in the strongly interacting quark-gluon plasma", I. Grishmanovskii, T. Song, **O. Soloveva**, C. Greiner and E. Bratkovskaya,
Phys. Rev. C, **106** (2022), 014903; e-print arXiv:2204.01561 [nucl-th];
- 15. "Soft gluon emission from heavy quark scattering in strongly interacting quark-gluon plasma", I. Grishmanovskii, T. Song, **O. Soloveva**,
Phys. Rev. D, **107** (2023), 036009; e-print arXiv:2210.04010 [nucl-th].
- 16. "Inelastic and elastic parton scatterings in the strongly interacting quark-gluon plasma", I. Grishmanovskii, **O. Soloveva**, T. Song, C. Greiner and E. Bratkovskaya,
Phys. Rev. C **109** (2024) 2, 024911, e-print arXiv:2308.03105 [hep-ph];
- 17. "Extraction of the microscopic properties of quasi-particles using deep neural networks",
O. Soloveva, A. Palermo, and E. Bratkovskaya,
Phys. Rev. C **110** (2024) 3, 034908, e-print arXiv:2311.15984[hep-ph];

- 18. “Jet transport coefficients by elastic and radiative scatterings in the strongly interacting quark-gluon plasma”,
I. Grishmanovskii, **O. Soloveva**, T. Song, C. Greiner and E. Bratkovskaya,
Phys. Rev. C 110 (2024) 1, 014908, e-print arXiv:2402.04923 [hep-ph];
- 19. “Finite density QCD equation of state: Critical point and lattice-based T’ expansion”,
M. Kahangirwe, S. A. Bass, E. Bratkovskaya, J. Jahan, P. Moreau, P. Parotto,
D. Price, C. Ratti, **O. Soloveva** and M. Stephanov,
Phys. Rev. D 109 (2024) no.9, 094046, e-print 2402.08636 [nucl-th];
- 20. “Thermal production of charm quarks in relativistic heavy-ion collisions”,
T. Song, I. Grishmanovskii, **O. Soloveva** and E. Bratkovskaya,
Phys. Rev. C 110 (2024) 3, 034906, e-print 2404.00425 [nucl-th];
- 21. “Influence of electromagnetic fields on the generation of the directed and elliptic flows of heavy quarks in relativistic heavy-ion collisions”,
S. K. Das, **O. Soloveva**, T. Song and E. Bratkovskaya,
Phys. Rev. C 112 (2025) 6, 064901, e-print arXiv:2507.22620 [nucl-th];
- 22. “CNN-Based Online Trigger for QGP Event Selection”,
O. Soloveva, A. Belousov, I. Kisel and E. Bratkovskaya,
Phys. Rev. C under review, e-print arXiv:2605.26280 [nucl-th];
- 23. “Transport coefficients of strongly interacting quark-gluon plasma including elastic and inelastic scattering within the dynamical quasiparticle model”,
G. Ingrosso, **O. Soloveva**, I. Grishmanovskii and E. Bratkovskaya,
Phys. Rev. D under review, e-print arXiv:2606.13363 [hep-ph].

Awards and Prizes

- 2019 Giersch-Excellence-Award for outstanding scientific work during the PhD studies
- 2019 Best theoretical poster at The XVIII International Conference on Strangeness in Quark Matter (SQM 2019) (young scientist prize sponsored by the Nuclear Physics European Collaboration Committee)
- 2014 — Winner of the competition for ITEP young scientists, engineers and technical workers
- 2017
- 2015 Winner of the competition “ NRNU MEPhI best student - 2015”
- 2016 Laureate of the “2nd ITEP premium of the competition in the scientific research (Theoretical works), dedicated to the 70th anniversary of ITEP “. Topic: “ Internal structure of mesons in the strong magnetic field”.

Selected talks

- May 18 - 20, 2026. Workshop “Heavy Quarks in the Quark-Gluon Plasma”, Bielefeld, Germany.
Talk title: “Jets and heavy flavor in strongly interacting QCD matter” ([pdf](#)).
- September 28 - October 4, 2023. STRONG NA7 Workshop and HFHF Theory Retreat, Giardini-Naxos, Italy.
Talk title: “Transport properties of the QGP matter facilitated by Machine learning-based models” ([pdf](#)).
- February 13 - 17, 2023. 4th International Workshop on QCD Challenges from pp to AA collisions, Padova, Italy
Talk title: “Soft gluon emission and jet quenching in the strongly interacting QGP” .

- Mar 28 - Apr 1, 2022. Mini Workshop "Phase transitions in particle physics", The Galileo Galilei Institute.
Talk title: "Influence of a phase transition on the transport properties of QCD matter" ([video](#)).
- Apr 4 - 10, 2022. The XXIXth International Conference on Ultra-relativistic Nucleus-Nucleus Collisions (Quark Matter 2022) (online).
Poster talk title: "Transport properties of the QGP in the dynamical quasi-particle model with a CEP" ([pdf](#)).
- Jun 13 - 18, 2022. The 20th International Conference on Strangeness in Quark Matter (SQM 2022), Busan, Republic of Korea.
Talk title: "Exploration of the phase diagram within a transport approach" ([pdf](#)).
- Sep 3 - 9, 2021. Workshop of the Network NA7-Hf-QGP of the European program "STRONG-2020".
Talk title: "Evolution of transport coefficients of the hot and dense QGP along the phase transition".
- May 20 - 25, 2019. the FAIR next generation scientists - 6th Edition Workshop (FAIRNESS2019), Genova, Italy.
Talk title: "Exploring the partonic phase at finite chemical potential within a selfconsistent off-shell transport approach" ([pdf](#)).
- Jun 20-25. The XVIII International Conference on Strangeness in Quark Matter (SQM 2019), Bari, Italy.
Poster title: "Transport coefficients of the hot and dense matter".
- Jun 18 - 24, 2017. 35th International Symposium on Lattice Field Theory (Lattice 2017), Granada, Spain.
Oral talk title: "Tetraquark and molecule charmonium states on the lattice".
- Aug 17 - 29, 2017. 6th International Conference on New Frontiers in Physics, Crete, Greece.
Talk title: "Charmed states on the lattice".
- Jul 24 - 30, 2016. 34st International Symposium on Lattice Field Theory (Lattice 2016), Southampton, The United Kingdom.
Talk title: "Lattice simulations of vector mesons in strong magnetic field".

Public talks and Seminars

- 08 May, 2023
"Dynamical properties of the QGP matter at finite temperatures and chemical potentials", The MUSES Collaboration seminar Series, University of Houston/Illinois/Kent, ([online](#)).
- 20 April, 2023
"Machine Learning for the extraction of the QGP properties", theory seminar at the INFN Sezione di Catania (INFN CT), Catania, Italy.
- 30 March, 2021
"Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition", Joint UH/UIUC/KSU Weekly journal club, University of Houston, (online).
- 30 March, 2021
"Shear viscosity and electric conductivity of a hot and dense QGP with a chiral phase transition", Joint UH/UIUC/KSU Weekly journal club, University of Houston, (online).

- 10 December, 2020
“Transport coefficients of the dense QGP along the chiral phase transition”, Transport meeting, ITP Goethe University, Frankfurt am Main, (online).
- 22 June, 2020
“QGP dynamics at finite chemical potential”, IGFAE Theory Seminars at Instituto Galego de Física de Altas Enerxías / Universidade de Santiago de Compostela (online) ([video](#)).
- 22 January, 2020
“Transport properties of the hot and dense QGP”, Seminar High Energy Physics, Bielefeld University, Germany.
- 8 July, 2019
“ μ_B -dependence of the QGP dynamics”, Palaver at ITP Goethe University, Frankfurt am Main, Germany.
- 14 September, 2017
“Mesons in strong magnetic field on the lattice”, Theory group Literary Seminar at ITEP (Institute for Theoretical and Experimental Physics), NRC Kurchatov Institute, Moscow, Russia.
- 11 December, 2017
“Light vector mesons in external magnetic field by Quenched SU(3) Lattice QCD”, Lattice Seminar at DESY Zeuthen, Germany.

Outreach

- 2017
Public talk “Early Universe and Fundamental Interactions”, Cultural and educational project “Kurilka Gutenberga”, Moscow, Russia. ([Youtube](#))
- 2015 - 2016
Organizer of (Quiz: theoretical physics in everyday life) Open faculty days for high school students, NRNU MEPhI, Moscow, Russia.

References

- Prof. Dr. E. Bratkovskaya
- Prof. Dr. J. Aichelin

Frankfurt am Main,
June 17, 2026

Olga Soloveva